WORKED SOLUTION

STEP III 2007, Question 8

Pure • Riccati Equations

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s3-q8

The key idea

The nonlinear equation $(*)$ is a Riccati equation: a y^2 term spoils linearity. Writing y as a multiple of u'/u turns $y' + y^2$ into u''/u, and that gets rid of the square, leaving a *linear* second-order equation for u. The first half of (i) hands us two solutions of exactly that linear equation, so everything else is fitting constants. For (ii) we need two solutions of a new linear equation, and the clue is that $1 - x$ is $1 + x$ with x replaced by $-x$.

Solution

Part (i)

Finding a **and** b**.** For $u = x$ we have $u' = 1$, $u'' = 0$, so we need

$$\mathrm{a}(x) + x\,\mathrm{b}(x) = 0.$$

For $u = \mathrm{e}^{-x}$ we have $u' = -\mathrm{e}^{-x}$, $u'' = \mathrm{e}^{-x}$; dividing by $\mathrm{e}^{-x} \neq 0$ gives

$$1 - \mathrm{a}(x) + \mathrm{b}(x) = 0.$$

The first equation gives $\mathrm{b} = -\mathrm{a}/x$ (for $x \neq 0$), and then the second gives $\mathrm{a}(1 + \frac{1}{x}) = 1$. So

$$\mathrm{a}(x) = \frac{x}{1+x}, \qquad \mathrm{b}(x) = -\frac{1}{1+x}.$$

(These also satisfy both conditions at $x = 0$.) The equation is linear and homogeneous, and x and e^{-x} are not multiples of each other, so the general solution is

$$u = Ax + B\mathrm{e}^{-x},$$

with A and B arbitrary constants.

The substitution. Put $y = \dfrac{u'}{3u}$, where $u \neq 0$. By the quotient rule,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{u''}{3u} - \frac{(u')^2}{3u^2} = \frac{u''}{3u} - 3\left(\frac{u'}{3u}\right)^2 = \frac{u''}{3u} - 3y^2.$$

So $y' + 3y^2 = \dfrac{u''}{3u}$: the square cancels exactly, which is why the factor 3 was put in the substitution. Equation $(*)$ becomes

$$\frac{u''}{3u} + \frac{x}{1+x} \cdot \frac{u'}{3u} = \frac{1}{3(1+x)},$$

and multiplying by $3u$ gives

$$\frac{\mathrm{d}^2u}{\mathrm{d}x^2} + \frac{x}{1+x}\frac{\mathrm{d}u}{\mathrm{d}x} - \frac{1}{1+x}u = 0,$$

as required. This is the equation from the start of the part, with the a and b we found.

The particular solution. So $u = Ax + B\mathrm{e}^{-x}$, and

$$y = \frac{u'}{3u} = \frac{A - B\mathrm{e}^{-x}}{3(Ax + B\mathrm{e}^{-x})}.$$

Ask yourself There are two constants but only one condition. Is that a problem?

No. Multiplying u by a non-zero constant doesn't change u'/u, so only the ratio $A : B$ matters, and a first-order equation should have just one free constant. The condition $y = 0$ at $x = 0$ gives $A - B = 0$. Then $B = A \neq 0$ (otherwise $u \equiv 0$), and cancelling A,

$$y = \frac{1 - \mathrm{e}^{-x}}{3(x + \mathrm{e}^{-x})}.$$

The denominator never vanishes, since $\mathrm{e}^{-x} \geq 1 - x$ gives $x + \mathrm{e}^{-x} \geq 1$. That completes part (i).

Part (ii)

Now the y^2 term has coefficient 1, so we use $y = \dfrac{u'}{u}$. Exactly as before,

$$y' = \frac{u''}{u} - \left(\frac{u'}{u}\right)^2 = \frac{u''}{u} - y^2,$$

so $y' + y^2 = u''/u$. Multiplying the equation by u gives the linear equation

$$u'' + \frac{x}{1-x}u' - \frac{1}{1-x}u = 0.$$

This is the equation of part (i) with x replaced by $-x$. Indeed, if $v(x) = w(-x)$ where w solves the part (i) equation, then $v'(x) = -w'(-x)$ and $v''(x) = w''(-x)$, and the part (i) equation at the point $-x$ reads

$$v''(x) + \frac{-x}{1-x}(-v'(x)) - \frac{1}{1-x}v(x) = 0,$$

which is the new equation. So from x and e^{-x} we get the solutions $-x$ (or just x) and e^{x}. Checking directly: for $u = x$, $0 + \frac{x}{1-x} - \frac{x}{1-x} = 0$; for $u = \mathrm{e}^{x}$, $\mathrm{e}^{x}(1 + \frac{x}{1-x} - \frac{1}{1-x}) = \mathrm{e}^{x}(1-1) = 0$. Hence

$$u = Ax + B\mathrm{e}^{x}, \qquad y = \frac{u'}{u} = \frac{A + B\mathrm{e}^{x}}{Ax + B\mathrm{e}^{x}}.$$

At $x = 0$ this is $(A+B)/B$, which must equal 2, so $A = B$ (and $B \neq 0$). Therefore

$$y = \frac{1 + \mathrm{e}^{x}}{x + \mathrm{e}^{x}}.$$

This is valid on the interval containing 0 on which $x + \mathrm{e}^{x} > 0$ and the equation is defined, namely $x_0 < x < 1$, where $x_0 \approx -0.567$ is the root of $x + \mathrm{e}^{x} = 0$.

Checks

- At $x = 0$ the part (ii) answer gives $2/1 = 2$, and differentiating it gives $y'(0) = \frac{1\cdot 1 - 2\cdot 2}{1} = -3$, which matches the equation: $1 - 4 - 0 = -3$.
- As $x \to \infty$ the part (i) answer tends to 0 like $\frac{1}{3x}$. Then $y' \approx -\frac{1}{3x^2}$ and $3y^2 \approx \frac{1}{3x^2}$ cancel, and $\frac{x}{1+x}y \approx \frac{1}{3(1+x)}$, consistent with $(*)$.
- We solved the part (ii) equation numerically (Runge–Kutta from $y(0) = 2$) and got $y(0.9) = 1.0297654\ldots$, agreeing with the formula to 10 decimal places.

Common slips

- Keeping the factor $\frac{1}{3}$ in part (ii): the multiplier must match the coefficient of y^2, so here it is $y = u'/u$.
- Trying to fix both A and B from one condition. Only their ratio affects y.
- Taking e^{-x} again in part (ii) without checking it: the sign change in $1 - x$ turns it into e^{x}.

Beyond STEP

An equation $y' + P(x)y^2 + Q(x)y = R(x)$ is a Riccati equation, and the substitution $y = u'/(Pu)$ always turns it into a linear second-order equation. Conversely, the ratio u'/u of any second-order linear equation satisfies a Riccati equation, which is why these nonlinear equations keep appearing in physics, from quantum mechanics to control theory. Boyce and DiPrima, *Elementary Differential Equations*, covers this and the theory of linear second-order equations (why two independent solutions give the general solution).

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